

Ligjerata 7dh

Shpejtësia e parë, e dytë dhe e tretë kozmike

$$F = \gamma \frac{mM}{r^2}$$

$$F_c = \frac{mv_1^2}{r}$$

$$F = F_c$$

$$\gamma \frac{mM}{r^2} = \frac{mv_1^2}{r}$$

$$v_1^2 = \frac{\gamma M}{r} \quad / \sqrt{\quad}$$

$$v_1 = \sqrt{\frac{\gamma M}{r}}$$

$$v_1 = \sqrt{\frac{g_0 R^2}{r}}$$

$$v_1 = \sqrt{g_0 \cdot R} = \sqrt{9.81 \frac{m}{s^2} \cdot 6.378 \cdot 10^6 m} = 7.9 \cdot 10^3 \frac{m}{s}$$

$$v_1 = 7.9 \frac{km}{s}$$

$$Q = mg^0; F = \gamma \frac{mM_T}{R_T^2}$$

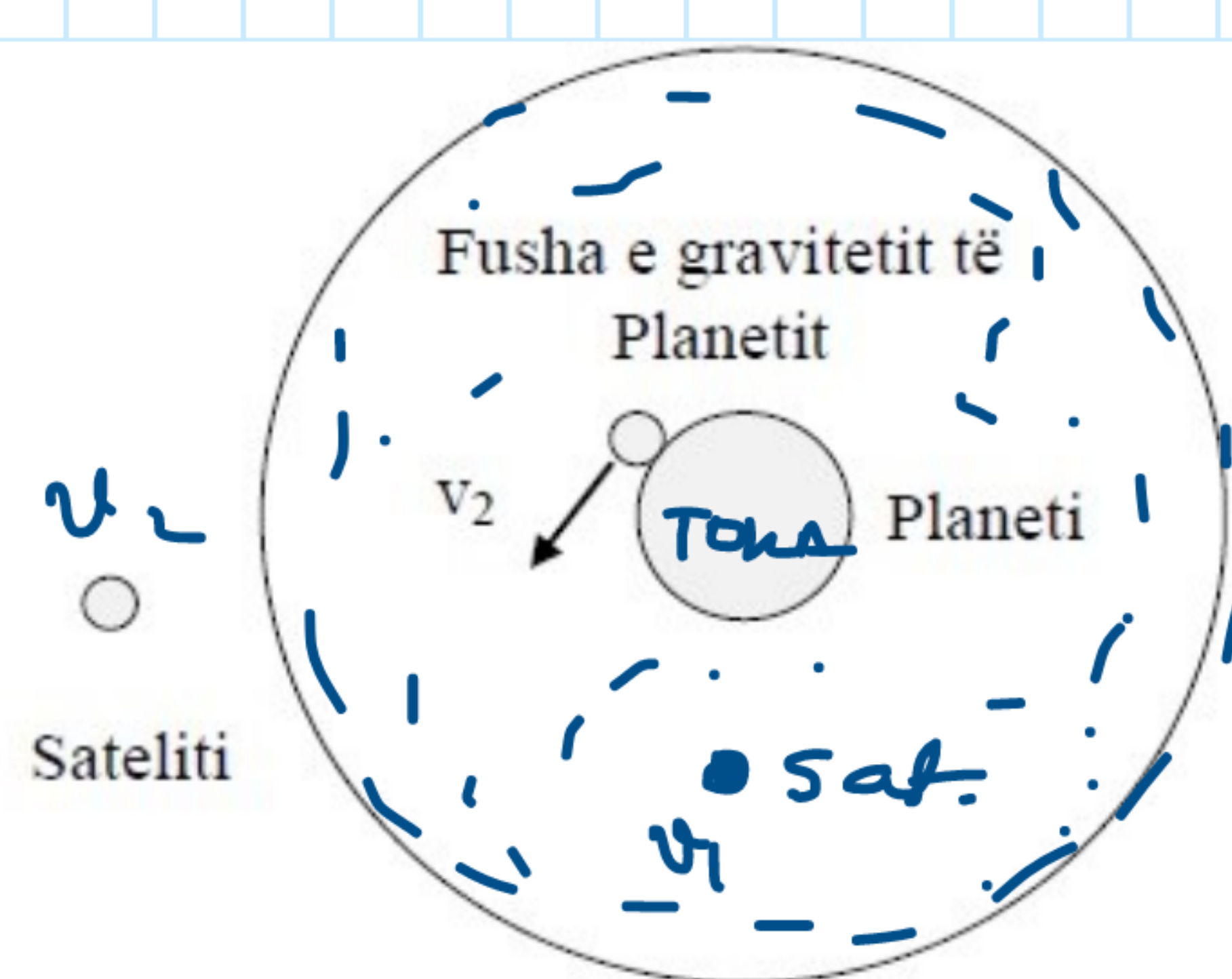
$$mg^0 = \gamma \frac{mM_T}{R_T^2}$$

$$g^0 \cdot R_T^2 = \gamma M_T$$

$$g^0 R^2 = \gamma M$$

$$R = 6.378 \cdot 10^6 m$$

$$g^0 = 9.81 \frac{m}{s^2}$$



Sh. II kozmike

$$E_{k_0} = \frac{mv_2^2}{2}; U = \gamma \frac{M}{r} m$$

$$E_{k_0} = U$$

$$\frac{mv_2^2}{2} = \gamma \frac{M}{r} m \quad \gamma M = gR^2$$

$$\frac{v_2^2}{2} = \frac{gR^2}{r} = gR \quad / \cdot 2$$

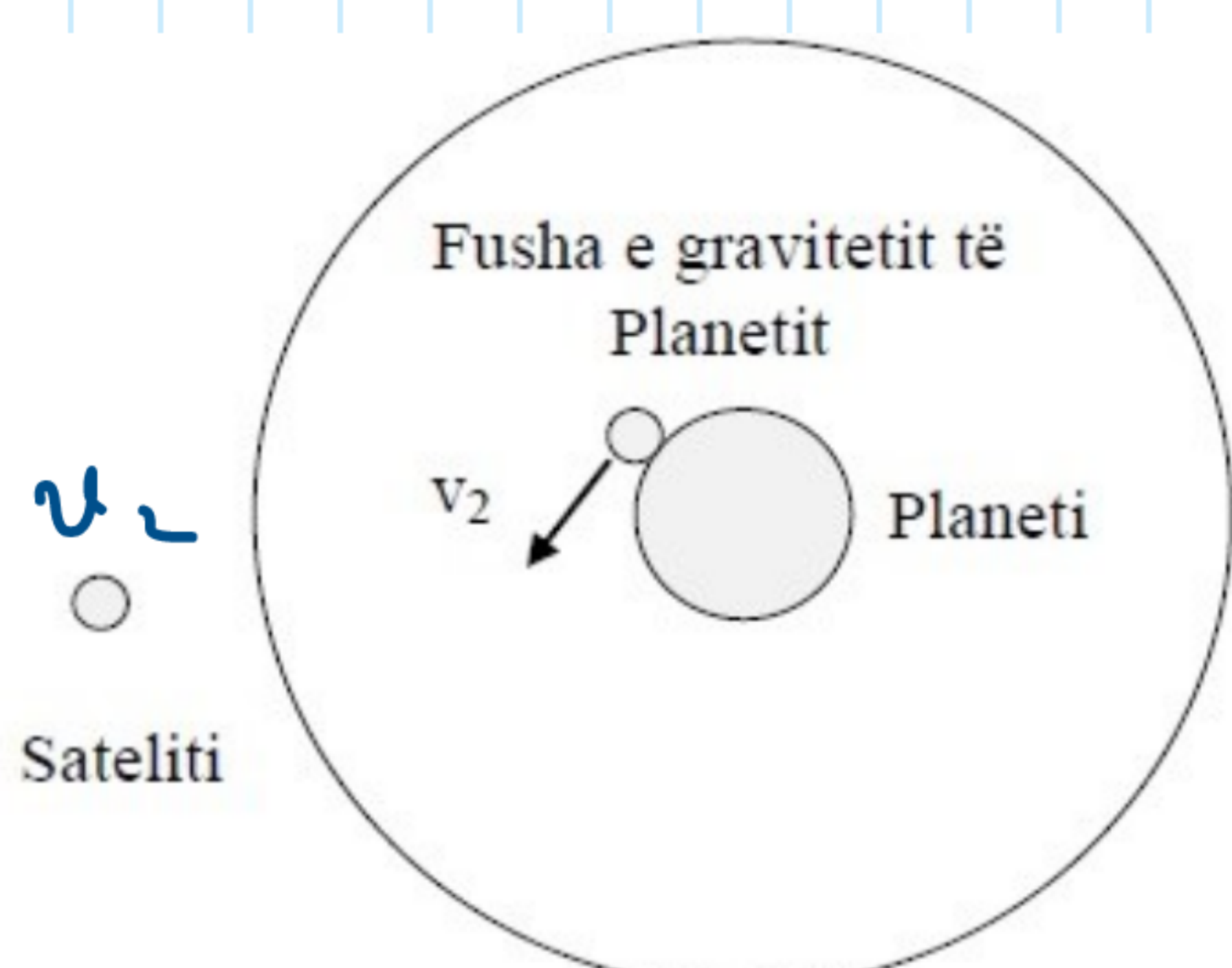
$$v_2^2 = 2gR \quad / \sqrt{\quad}$$

$$v_2 = \sqrt{2gR} = \sqrt{2} \cdot \underbrace{\sqrt{gR}}_{=v_1}$$

$$v_2 = \sqrt{2} \cdot v_1$$

$$v_2 = 1.41 \cdot v_1 = 1.41 \cdot 7.9 \cdot 10^3 \frac{m}{s} = 11.1 \cdot 10^3 \frac{m}{s}$$

$$v_2 = 11.1 \frac{km}{s}$$



Sh. III kozmike

$$E_{k_0} = \frac{mv_3^2}{2}; U_b = \gamma \frac{mM_D}{R}$$

$$\frac{mv_3^2}{2} = \gamma \frac{mM_D}{R}$$

$$v_3^2 = \frac{2\gamma M_D}{R} \quad / \sqrt{\quad}$$

$$v_3 = \sqrt{2\gamma \frac{M_D}{R}}$$

$$M_D = 2 \cdot 10^{30} kg$$

$$R = 1.5 \cdot 10^8 km \quad (\text{distanca Tokë-Dielli})$$

$$\gamma = 6.67 \cdot 10^{-11} \frac{Nm^2}{kg^2}$$

$$v_3 = 42.2 \frac{km}{s}$$

$$Sh. IV kozmike > 285 \frac{km}{s}$$